SmartCricket AI

Presentation guide: the project explained step by step, and what to say on every slide

This guide is for you, not for the staff. Read it once fully, then use Part 4 while you practise with the deck **SmartCricket_AI_Project_Review.pptx**. The same words are also in the speaker notes under each slide.

Part	What it covers
1	The project in simple words
2	How it works, step by step
3	Words you must be able to explain
4	Slide-by-slide guide: what each slide shows and what to say
5	Live demo steps
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Part 1. The project in simple words

Say it in one breath

SmartCricket AI is a web app. A batter uploads a photo or a short video. The app finds the body, measures the joint angles, names the shot, shows why it decided that, compares the batter with a good player, and an AI coach explains everything in plain words and gives a daily training plan.

The problem

- Cricket shots are still tagged by hand. Someone watches every ball and types the shot name. It is slow.
- Club and college players rarely have a coach at every session, so they get little feedback.

The base paper

- **CricShot10k** (Dihan et al., IEEE Access, 2026) built a large dataset: 10,086 one-second clips of 15 shot types.
- It also built **CricShotNet**: crop the batter, take 15 frames, EfficientNetV2-S reads each frame, a GRU follows the swing, and it names the shot. It reports 89% accuracy.
- But it only gives a label. No explanation, no app, no photo support, no body pose, no coaching.

What you added (your contribution)

Base paper has	Your project adds
Video model only	Video model + body pose model + fused model, compared fairly on the same test clips
A label and a number	Calibrated confidence, top-3 answers, heatmaps, key-frame timeline, 'uncertain' flag
No photo support	Photo posture report: quality check, 33 body points, joint angles in 2D and 3D
No comparison	Compare with a model player using DTW, joint by joint
No app	Full web app: player and admin roles, history, PDF reports, dashboard
No feedback	AI coach (text + voice), daily training coach, live camera mirror, phone mode

The research question and the answer

Question: does adding body pose (the skeleton) to the video model help it recognise shots?

Answer: yes. Video only: 56.2% accuracy. Video + pose: 63.9%. That is +7.8 points, and the McNemar test shows it is not luck ($p = 4.3 \times 10^{-9}$).

Part 2. How it works, step by step

A. Photo mode (posture report)

- 1. The player uploads a JPG or PNG photo.
- 2. **Quality check:** the app checks sharpness, light, body size and camera angle, and gives a tip for each problem.
- 3. **MediaPipe** finds 33 body points (shoulders, elbows, knees and so on) with a visibility score for each.
- 4. **Joint angles** are measured from three points, for example hip-knee-ankle gives the knee angle. Unclear points are greyed out, never guessed.
- 5. A 3D view shows the skeleton from any side.
- 6. The report (picture + table) is saved, and the AI coach can explain it.
- Note: a photo does not give a shot name, because one frame cannot show a movement.

B. Video mode (shot analysis)

- 1. The player uploads a short MP4, MOV or AVI clip.

- 2. The app checks the file, **follows the batter** (even if the bowler or keeper is in view) and picks **15 frames**.
- 3. Two streams: **EfficientNetV2-S + GRU** reads the pictures; **MediaPipe** gives the pose (angles and positions) for each frame.
- 4. The **fused model** joins both and gives one of 15 shots, with top-3 answers.
- 5. **Temperature scaling** makes the confidence honest. Low-confidence clips are marked 'uncertain'.
- 6. **Grad-CAM heatmap** shows where the model looked; the **key-frame timeline** shows which moment mattered most.
- 7. Optional: **compare with a reference clip** of a good player using DTW.
- If no trained model is installed, the app still gives a **movement analysis** (body tracked, joint ranges, busiest frame).

C. AI Coach

- Only the stored numbers (shot, confidence, angles) are sent to the language model (Gemini), never the photo or video.
- Each line starts with 'Measured:' (from the analysis) or 'General tip:' (coaching knowledge).
- A software check rejects any reply that contains a number that was not measured. It asks again, then falls back safely.
- It can read answers aloud (male voice) and take spoken questions.

D. Training coach, live mirror and the rest

- **Daily plan:** three drills a day from the player's weak joints; a morning check-in (sleep, soreness) makes it easier or harder.
- **Motivation:** levels (Net Rookie to Pro), XP, streaks, a daily quiz and a weekly report written by the coach.
- **Live mirror:** the camera shows a live skeleton at 30-60 frames per second on the device; it counts reps, speaks corrections and obeys voice commands. 'Shadow the Pro' shows a gold ghost of a good player.
- **Phone mode:** scan a QR code to use the phone camera at the nets (secure https on the home Wi-Fi).
- **History and reports:** every result saved, PDF and CSV download. **Admin:** players, reference clips, model results.

E. How the models were trained

- Training ran in a Google Colab notebook on a free T4 GPU, using CricShot10k.

- Split: 7,304 training, 1,498 validation and 1,289 test clips. No test clip was used for training or choosing settings.
- Five models were trained: video only, pose only (2D), video + pose (2D), pose only (3D), video + pose (3D).
- The trained files (video.pt, pose.pt, fusion.pt, pose3d.pt, fusion3d.pt) are in the app's models folder.

Part 3. Words you must be able to explain

Word	Simple meaning
CNN / EfficientNetV2-S	A network that looks at one picture and turns it into numbers describing what is in it. EfficientNetV2-S is a fast, accurate one, pre-trained on ImageNet.
GRU	A network that reads a sequence (the 15 frames) in order, so it understands movement over time.
Attention	Gives each of the 15 frames a weight; the highest weight is the key frame.
MediaPipe Pose	Google's tool that finds 33 body points in a picture, with a visibility score for each.
Joint angle	The angle at a joint from three points, e.g. hip-knee-ankle for the knee.
2D vs 3D pose	2D angles are measured on the picture and change with camera angle; 3D uses MediaPipe's estimated real-world positions.
Fusion	Joining the video features and the pose features before the final answer.
Grad-CAM	A heatmap showing which parts of the frame pushed the model to its answer. It shows where it looked, not proof that it reasoned correctly.
Temperature scaling (calibration)	One number T that softens or sharpens the scores so that '80% sure' really means right about 80% of the time.
ECE	Expected calibration error: how far the confidence is from the real accuracy. Lower is better.
DTW	Dynamic Time Warping: lines up two movements even if one is faster, then compares them frame by frame.
Top-1 / Top-3 accuracy	Top-1: the first answer is right. Top-3: the right shot is among the first three answers.
Macro-F1	The average F1 score over all 15 shots, so rare shots count as much as common ones.
McNemar test	A statistical test for two models on the same clips. It looks only at clips where they disagree.

Word	Simple meaning
FastAPI / PostgreSQL	FastAPI is the Python back end (the API). PostgreSQL is the database that stores users and results.
JWT / roles	A signed token given at sign-in. It says who you are and whether you are a player or an admin.

Part 4. Slide-by-slide guide

About 1 to 1.5 minutes per slide, 25 slides, so plan for 25 to 30 minutes plus questions and demo. The italic text is what to say; the same text is in the speaker notes.

1. Title

Shows: Project name, base paper, your details.
Say: Good morning. My project is SmartCricket AI. It is a web app that looks at a photo or a short video of a batter, tells them which shot they played and how their body moved, and coaches them. It is built on the CricShot10k paper from IEEE Access, 2026.

2. Abstract

Shows: Four cards: problem, starting point, what I built, main result. Photo with the skeleton drawn by the app.
Say: This slide is the whole project. Shots are tagged by hand. The base paper gave a big dataset and a model. I built a full app around it and tested whether adding the body skeleton helps. It does: accuracy rose from 56% to 64%.

3. Why this project

Shows: Four reasons.
Say: Manual tagging is slow, most players have no coach every day, other sports already use AI, so I made a proven model explain itself and put it in an app.

4. Objectives

Shows: Five objectives, all marked DONE.
Say: Rebuild the base model, add pose, compare the models, build photo and video modes, and explain and coach. All five are complete.

5. Literature survey

Shows: Table of four works and a bar chart of dataset sizes.
Say: Earlier datasets are small. Rao et al. used only pose, which misses the bat. CricShot10k is five times bigger, so I chose it.

6. Existing system

Shows: CricShotNet pipeline, crop and tint pictures, the authors' 89% result.
Say: Crop the batter, take 15 frames, tint bat and batter, EfficientNet reads frames, GRU follows the swing. These are the authors' numbers.

7. Disadvantages

Shows: Six gaps.
Say: No explanation, no app, only TV footage, no pose, no photos, confidence not checked. These are gaps for my goal, not mistakes in their paper.

8. Proposed system

Shows: Photo flow, video flow, and three extras.
Say: Photo mode gives a posture report. Video mode names the shot, explains it and compares with a model player. On top: AI coach, daily training and a live mirror.

9. All features

Shows: Twelve features in a grid.
Say: These are the twelve features of the finished app. Every screen I show next is real, not a mock-up.

10. Advantages

Shows: Six advantages.
Say: Photo or video, answers you can check, flags when unsure, plain-language coach, private live mirror, low cost.

11. Architecture

Shows: Layers from browser to database.
Say: Read top to bottom: browser, FastAPI, OpenCV preprocessing, then the shot classifier and pose analysis, then database, coach and reports.

12. Software

Shows: Five columns of tools.
Say: Plain HTML, CSS and JavaScript; FastAPI and PostgreSQL; PyTorch; OpenCV and MediaPipe; Colab for training, and 64 automated tests.

13. M1 Interface

Shows: Dashboard screenshot.

Say: Two roles with separate pages. After sign-in the player sees the scoreboard, today's plan, batting profile and best photo.

14. M2 Photo report

Shows: Posture report screenshot.

Say: Quality check first, then 33 points and the joint angles. Unclear points are greyed out. No shot guess from a photo.

15. M3-M4 Video

Shows: Shot analysis screenshot (Cover Drive 98%).

Say: The app follows the batter, picks 15 frames and the fused model names the shot. The heatmap shows where it looked; the timeline shows the key moment.

16. M5 Compare

Shows: Comparison card.

Say: DTW lines up my swing with a model player's and shows the angle gap at every joint, for example 44 degrees at the right shoulder.

17. M6-M7 AI Coach

Shows: Coach chat screenshot.

Say: The coach only sees the numbers. Every line is Measured or General tip, and a check rejects made-up numbers. It can also speak.

18. M9 Training

Shows: Training page screenshot.

Say: Three drills a day from my weak joints; the check-in changes the difficulty; levels, XP, streaks, quiz and a weekly report.

19. Live mirror

Shows: Live skeleton screenshot.

Say: Live skeleton on the device, rep counting, voice commands, Shadow the Pro, and phone mode with a QR code.

20. M8 Admin

Shows: Admin overview screenshot.
Say: History and PDF for players. The admin manages players, reference clips and sees model results.

21. Results 1

Shows: Bar chart of five models, three big numbers.
Say: Video only 56%, pose only 50%, video + pose 64%. +7.8 points, and McNemar says it is not luck. The base paper's 89% used YOLO cropping and full fine-tuning, so my numbers are lower; my question was whether pose helps, and it does.

22. Results 2

Shows: Six result cards.
Say: Calibration made the confidence honest; pose helped most for front-on clips; shoulders and hips are found reliably; a photo takes under 50 ms and a video about 3 seconds on a laptop; all 64 tests pass.

23. Conclusion

Shows: Achievements and future work.
Say: I built the app, proved pose helps, and made the answers trustworthy. Next: YOLO cropping and full training, video transformers, more shots, a mobile app.

24. References

Shows: Ten references.
Say: Reference 1 is the base paper; the rest are the standard papers for each method.

25. Thank you

Shows: Questions and demo.
Say: Thank you. I can show a live demo now.

Part 5. Live demo steps

Start the app 10 minutes before the review and keep the browser open on the sign-in page.

- 1. Open Terminal and run:
  ```
  cd ~/Desktop/SmartCricket/smartcricket-ai
  .venv/bin/uvicorn backend.app.main:app --port 8000
  ```
- 2. Open **http://localhost:8000** in Chrome. Sign in with the player account from LOGIN_DETAILS.txt.

- 3. **Photo:** Photo analysis, upload a batting photo, show the quality strip, skeleton, angles and 3D view.
- 4. **Video:** Video analysis, upload a clip from sample_clips, show the shot, top-3, heatmap and timeline. Then pick a reference to compare.
- 5. **Coach:** ask 'Why is this a cover drive?' and press the speaker button to hear it.
- 6. **Training:** show today's three drills, the check-in and the level.
- 7. **Live mirror:** Start camera, stand side-on, show the skeleton and rep counter. Say 'stop' to stop.
- 8. **Admin:** sign out, sign in as admin, show the overview and Model results.
- Keep the demo under 5 minutes. If the internet is slow, the coach may take a few seconds; everything else works offline.

Part 6. Questions staff may ask

Why is your accuracy (64%) lower than the paper's 89%?
The paper crops the batter with YOLO and fine-tunes the whole network. I kept the EfficientNet backbone frozen and used no YOLO cropping, to train on a free Colab GPU in a few hours. My goal was to test whether pose helps under the same conditions, and it gave +7.8 points. Cropping and full fine-tuning are my future work.

What is new in your project compared with the base paper?
Pose fusion with a fair three-way comparison, calibrated confidence, heatmaps and key frames, photo mode, reference comparison with DTW, the grounded AI coach, the training coach, the live mirror and the full web app.

How do you know the improvement is real?
All models were tested on the same 1,289 clips, and McNemar's exact test gives p = 4.3 x 10^-9, so the difference is not luck.

Why not predict the shot from a photo?
A shot is a movement. One frame cannot show the bat swing, so the base paper and I use video for shots and give posture only for photos.

Can the AI coach make things up?
It only receives the measured numbers. Every number in its reply is checked against them; if one is not found it is asked again, and then a safe fallback message is shown.

What does temperature scaling do?

It adjusts the scores with one number T so that the confidence matches real accuracy. It lowered the calibration error by 22% to 44% without changing any prediction.

What is DTW and why use it?

Dynamic Time Warping lines up two movements even if one is faster, so a slow player can be compared with a fast pro frame by frame.

Did 3D pose help?

Not for classification: the difference from 2D was not significant, because the 3D points are estimated from the same 2D picture. 3D is still useful for measuring angles independent of the camera.

Where does the data come from? Is it allowed?

CricShot10k is public under CC BY 4.0. Photos in the report and slides are from Wikimedia Commons with credits.

How is user data protected?

Passwords are hashed, sign-in uses tokens with player and admin roles, the live mirror runs on the device, and only numbers are sent to the language model.

How long does analysis take?

On a laptop CPU: photo under 50 ms, movement analysis about 0.25 s, full video shot analysis about 3 s, live mirror 30-60 frames per second.

How did you test it?

64 automated tests (unit, integration and system) all pass, plus the experiments E1 to E10 in the report.

What are the limitations?

Broadcast-style clips work best; side-on broadcast clips are hardest; 2D angles depend on camera angle; heatmaps are weak for some shots.

Part 7. Checklist before the review

- Fill in the college name and guide name on slides 1 and 25 (and the title page of the report).
- Open the deck in PowerPoint once: check Century Gothic shows correctly on the review computer.
- Practise once with a timer: aim for about 25 minutes.
- Start the app and run the demo once on the day; keep a sample photo and clip ready on the desktop.
- Charge the laptop; carry the report PDF and the deck on a pen drive as a backup.